

ALMIRAL GEL – PACKAGE INSERT**1. NAME OF THE MEDICINAL PRODUCT**

Almiral 1% gel

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 1 g of gel contains 10 mg diclofenac sodium as diclofenac diethylamine.

Excipient with known effect: propylene glycol.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Gel

Colourless, transparent, smooth, homogeneous gel with a characteristic peppermint odour.

4. CLINICAL PARTICULARS**4.1. Therapeutic indications**

For the external treatment of pain, inflammation and swelling in the following situations:

- injuries of the tendons, ligaments, muscles and joints, e.g. sprains, bruises, strains and back pain after sport or accidents
- localized forms of soft tissue rheumatism such as tendinitis (tennis elbow), shoulder hand syndrome, bursitis, periarthropathies
- for the symptomatic treatment of osteoarthritis of small and medium sized joints located close to the skin such as the finger joints or knee joints

4.2. Posology and method of administration

Posology

For cutaneous use only.

Patients should consult their doctor if the condition does not improve or worsens within 7 days of starting treatment.

For adults and adolescents aged 12 years and over.

This medicine should be applied over the affected area 3 or 4 times daily and rubbed gently into the skin. The amount needed depends on the size of the painful area: 2 g to 4 g Almiral gel (a quantity ranging in size from a cherry to a walnut) is sufficient to treat an area of about 400-800 cm². After application, the hands should be washed, unless they are the site being treated.

The duration of treatment depends on the indication and clinical response. The product should not be used for more than 14 days. Almiral gel can also be used as an adjunctive therapy with other pharmaceutical forms of Almiral gel.

Elderly patients (over 65 years of age)

The usual adult dosage may be used.

Children (under 12 years of age)

The use and safety of Almiral gel in children under 12 years systematically and is not recommended.

4.3. Contraindications

- Hypersensitivity to diclofenac or to any of the excipients listed in section 6.1.
- Patients in who attacks of asthma, urticaria or acute rhinitis are precipitated by acetylsalicylic acid (aspirin) or other non-steroidal anti-inflammatory drugs (NSAIDs).
- During the last trimester of pregnancy.
- Use in patients hypersensitive to propylene glycol or isopropanol or other components of the gel base.

4.4. Special warnings and precautions for use

The possibility of systemic adverse events from application of Almiral gel cannot be excluded if the preparation is used on large areas of skin and over a prolonged period (see the product information on systemic forms of diclofenac).

Discontinue the treatment if a skin rash develops after applying the product.

Almiral gel can be used with non-occlusive bandages but should not be used with an airtight occlusive dressing.

Almiral gel should be applied only to intact, non-diseased skin and not to skin wounds or open injuries. It should not be allowed to come into contact with the eyes or mucous membranes, and should not be ingested.

Side effects include itching, reddening or smarting of the skin or skin rash. Photosensitivity reactions have been observed in isolated cases.

Asthma has been rarely reported in patients using topical non-steroidal anti-inflammatory drugs (NSAID) preparations.

Application over extensive areas for prolonged periods or application in excess of recommended dosage may give rise to systemic effects. These include gastrointestinal disturbances and bleeding, irritability, fluid retention, rash, hepatitis, renal dysfunction, anaphylaxis and rarely blood dyscrasias, bronchospasm and erythema multiforme.

This product should only be used with great caution in patients with a history of peptic ulcer, gastrointestinal bleeding, hepatic or renal insufficiency, or bleeding diathesis, or intestinal inflammation. Circulating levels of the active drug substance are low but the theoretical risk in these patients should be considered.

Almiral gel contains propylene glycol, which may cause skin irritation.

4.5. Interactions with other medicinal products and other forms of interaction

Since systemic absorption of diclofenac from topical application is very low, such interactions are very unlikely.

However, the following interactions occur with oral forms of Almiral: *Lithium and digoxin*: Diclofenac may increase plasma concentrations of lithium and digoxin.

Anticoagulants: Although clinical investigations do not appear to indicate that diclofenac has an influence on the effect of anticoagulants, there are isolated reports of an increased risk of haemorrhage with the combined use of diclofenac and anticoagulant therapy. Therefore, to be certain that no change in anticoagulant dosage is required, close monitoring of such patients is required. As with other non-steroidal anti-inflammatory agents, diclofenac in a high dose can reversibly inhibit platelet aggregation.

Antidiabetic agents: Clinical studies have shown that diclofenac can be given together with oral antidiabetic agents without influencing their clinical effect. However there have been isolated reports of hypoglycaemic and hyperglycaemic effects which have required adjustment to the dosage of hypoglycaemic agents.

Cyclosporin: Cases of nephrotoxicity have been reported in patients receiving concomitant cyclosporin and NSAIDs, including diclofenac. This might be mediated through combined renal antiprostaglandin effects of both the NSAID and cyclosporin.

Methotrexate: Cases of serious toxicity have been reported when methotrexate and NSAIDs are given within 24 hours of each other. This interaction is mediated through accumulation of methotrexate resulting from impairment of renal excretion in the presence of the NSAID.

Quinolone antimicrobials: Convulsions may occur due to an interaction between quinolones and NSAIDs. This may occur in patients with or without a previous history of epilepsy or convulsions. Therefore, caution should be exercised when considering the use of a quinolone in patients who are already receiving an NSAID.

Other NSAIDs and steroids: Co-administration of diclofenac with other systemic NSAIDs and steroids may increase the frequency of unwanted effects. Concomitant therapy with aspirin lowers the plasma levels of each, although the clinical significance is unknown.

Diuretics: Various NSAIDs are liable to inhibit the activity of diuretics. Concomitant treatment with potassium-sparing diuretics may be associated with increased serum potassium levels, hence serum potassium should be monitored.

4.6. Fertility, pregnancy and lactation

Pregnancy

The systemic concentration of diclofenac is lower after topical administration, compared to oral formulations. With reference to experience from treatment with NSAIDs with systemic uptake, the following is recommended: Inhibition of prostaglandin synthesis may adversely affect the pregnancy and/or the embryo/fetal development. Data

from epidemiological studies suggest an increased risk of miscarriage and of cardiac malformation and gastroschisis after use of a prostaglandin synthesis inhibitor in early pregnancy. The absolute risk for cardiovascular malformation was increased from less than 1%, up to approximately 1.5%. The risk is believed to increase with dose and duration of therapy. In animals, administration of a prostaglandin synthesis inhibitor has been shown to result in increased pre- and post-implantation loss and embryo-fetal lethality. In addition, increased incidences of various malformations, including cardiovascular, have been reported in animals given a prostaglandin synthesis inhibitor during the organogenetic period. During the first and second trimester of pregnancy, diclofenac should not be given unless clearly necessary. If diclofenac is used by a woman attempting to conceive, or during the first and second trimester of pregnancy, the dose should be kept as low and duration of treatment as short as possible.

During the third trimester of pregnancy, all prostaglandin synthesis inhibitors may expose the fetus to:

- cardiopulmonary toxicity (with premature closure of the ductus arteriosus and pulmonary hypertension);
- renal dysfunction, which may progress to renal failure with oligo-hydroamniosis.

The mother and the neonate, at the end of pregnancy, to:

- possible prolongation of bleeding time, an anti-aggregating effect which may occur even at very low doses.
- inhibition of uterine contractions resulting in delayed or prolonged labour.

Consequently, diclofenac is contraindicated during the third trimester of pregnancy.

Breastfeeding

Like other NSAIDs, diclofenac passes into breast milk in small amounts. However, at therapeutic doses of diclofenac no effects on the suckling child are anticipated. Because of a lack of controlled studies in lactating women, the product should only be used during lactation under advice from a healthcare professional. Under this circumstance, Almira gel should not be applied on the breasts of nursing mothers, nor do elsewhere on large areas of skin or for a prolonged period of time (see section 4.4).

4.7. Effects on ability to drive and use machines

Patients who experience dizziness or other central nervous system disturbances, including visual disturbances, while taking NSAIDs should refrain from driving or operating machinery.

4.8. Undesirable effects

Table 1

<u>Infections and infestations:</u>	
Very rare:	Rash pustular.
<u>Immune system disorder:</u>	
Very rare:	Hypersensitivity (including urticaria), angioneurotic oedema.
<u>Respiratory, thoracic and mediastinal disorders:</u>	
Very rare:	Asthma.
<u>Skin and subcutaneous tissue disorders:</u>	
Common:	Rash, eczema, erythema, dermatitis (including dermatitis contact), pruritus
Rare:	Dermatitis bullous
Very rare:	Photosensitivity reaction

Almira gel is usually well tolerated. Itching, reddening or smarting of the skin, or skin rash, commonly occurs. Photosensitivity reactions have very rarely been observed.

Systemic absorption of Almira gel is low compared with plasma levels obtained following oral forms of Almira. However, where diclofenac gel is applied to a relatively large area of skin and over a prolonged period, the possibility of systemic side effects cannot be completely excluded.

Asthma has very rarely been reported in patients using topical NSAID preparations.

4.9. Overdose

Signs and symptoms

The low systemic absorption of topical diclofenac renders overdose very unlikely. However, undesirable effects, similar to those observed following an overdose of diclofenac tablets, can be expected if Almira gel is inadvertently ingested (1 tube of 100g contains the equivalent of 1000mg of diclofenac sodium). In the event of accidental ingestion, resulting in significant systemic adverse effects, general therapeutic measures normally adopted to treat poisoning with non-steroidal anti-inflammatory medicines should be used. Gastric decontamination and the use of activated charcoal should be considered, especially within a short time of ingestion.

Treatment

Management of overdosage with NSAIDs essentially consists of supportive and symptomatic measures. There is no typical clinical picture resulting from diclofenac overdosage. Supportive and symptomatic treatment should be given for complications such as hypotension, renal failure, convulsions, gastro-intestinal irritation, and respiratory depression; specific therapies such as forced diuresis, dialysis or haemoperfusion are probably of no help in eliminating NSAIDs due to their high rate of protein binding and extensive metabolism.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: Antiinflammatory preparations, non-steroids for topical use, Topical products for joint and muscular pain, ATC code: M02AA15.

Mechanism of action:

Diclofenac is a non-steroidal anti-inflammatory (NSAID) with pronounced analgesic, anti-inflammatory and antipyretic properties. Inhibition of prostaglandin synthesis is the primary mechanism of action of diclofenac.

Almiral gel is an anti-inflammatory and analgesic preparation designed for topical application. In inflammation and pain of traumatic or rheumatic origin, Almiral gel relieves pain, decreases swelling, and shortens the time to return to normal function.

Due to an aqueous-alcoholic base it exerts a soothing and cooling effect.

5.2. Pharmacokinetic properties

Absorption:

The quantity of diclofenac absorbed through the skin is proportional to the size of the treated area, and depends on both the total dose applied and the degree of skin hydration. Absorption amounts to about 6% of the applied dose of diclofenac after topical application of 2.5 g diclofenac gel on 500 cm² skin, determined by reference to the total renal elimination, compared with diclofenac tablets. A 10-hour occlusion leads to a three-fold increase in the amount of diclofenac absorbed.

Distribution:

Diclofenac concentrations have been measured from plasma, synovial tissue and synovial fluid after topical administration of diclofenac gel to hand and knee joints. Maximum plasma concentrations are approximately 100 times lower than after oral administration of the same quantity of diclofenac. 99.7% of diclofenac is bound to serum proteins, mainly albumin (99.4%).

Diclofenac accumulates in the skin which acts as reservoir from where there is a sustained release of drug into underlying tissues. From there, diclofenac preferentially distributes and persists in deep inflamed tissues, such as the joint, where it is found in concentrations up to 20 times higher than in plasma.

Biotransformation:

Biotransformation of diclofenac involves partly glucuronidation of the intact molecule, but mainly single and multiple hydroxylation resulting in several phenolic metabolites, most of which are converted to glucuronide conjugates. Two of the phenolic metabolites are biologically active, however, to a much smaller extent than diclofenac.

Elimination:

The total systemic clearance of diclofenac from plasma is 263 ± 56 ml/min. The terminal plasma half-life is 1-2 hours. Four of the metabolites, including the two active ones, also have short plasma half-lives of 1-3 hours. One metabolite, 3'-hydroxy-4'-methoxy-diclofenac, has a longer half-life but is virtually inactive. Diclofenac and its metabolites are excreted mainly in the urine.

Characteristics in patients:

No accumulation of diclofenac and its metabolites is to be expected in patients suffering from renal impairment. In patients with chronic hepatitis or non-decompensated cirrhosis, the kinetics and metabolism of diclofenac are the same as in patients without liver disease.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Propylene glycol

Isopropanol

Carboxypolymethylene (carbopol 940)

Dipropylene glycol monomethylether
Peppermint oil
Purified water

6.2. Incompatibilities

None stated.

6.3. Shelf life

Two (2) years

6.4. Special precautions for storage

Store below 30°C, in the original package.

6.5. Nature and contents of container

Lacquered aluminium tubes sealed with cap of 25g gel.

6.6. Special precautions for disposal and other handling

No special requirements.

7. PRODUCT REGISTRATION HOLDER

Komedic Sdn Bhd No. 4, Jalan PJS 11/14 Bandar Sunway 46150 Petaling Jaya Selangor Malaysia

8. MANUFACTURER

MEDOCHEMIE LTD (Cogols Facility), Konstantinoupoleos 1-10, Limassol, 3011, Cyprus

9. PRODUCT REGISTRATION NUMBER

MAL19913223AZ

10. DATE OF REVISION OF THE TEXT

05/2025